

Birthday word generator

Do you know which words entered the English language around the same time you entered the world? <http://dgit.in/bgantor>



Please don't be a hoax

Not very fast, and not the safest thing on the road but who wouldn't want a full sized lego car? <http://dgit.in/legocarr>

despite the complete lack of scientific scrutiny or peer review, he and his investors are taking advance orders of USD \$500,000 or more to sell these little magic boxes of energy. Even going so far as to invite pre-bookings for a portable 1MW unit for USD \$1.5 million in November 2013 with estimated delivery in early 2014.

Even as the world waits for the E-Cat to be tested the needs of the world don't change. With world energy requirements expected to hit a sharp increase from 524 quadrillion Btu (British thermal units) (i.e. 1.5356924×10^{14} kilowatt hours) in 2010 to 820 quadrillion Btu (i.e. $2.40318278 \times 10^{14}$ kilowatt hours) by 2040 it is clear that we need more power, cheaper, cleaner and safer.

on the other hand is a malicious fraud. And his biggest contribution may unfortunately be to further damage the image of cold fusion as a genuine science.

Reinventing Cold Fusion!

Perhaps the greatest irony in the history of cold fusion is the origin of its name. The term arose in 1956 and became popularly associated with the idea of self-sustained nuclear fusion but it was originally coined by journalist William Laurence writing about a different type of fusion (a far less sexy type) called the Muon-catalyzed fusion reaction.

Key feature of muon-catalyzed fusion is that although considered distinct from "cold fusion", in its popular sense it

problem in muon-catalyzed fusion). This field rechristened itself in the wake of "cold fusion's" bad publicity as Low Energy Nuclear Reactions (LENR) research and has continued its parallel quest towards achieving "cool fusion" (as we like to call it).

Cool fusion or LENR research may have the answer that cold fusion research seems to be too hard pressed or distracted to find. For the most part cold fusion and LENR are on the same journey of discovery but over a period of time they've attracted different sorts of researchers. The former is a litany of mistaken scientist (perhaps too proud to admit their errors) and charlatans who attempt to profit from selling snake oil, whereas the latter is a quiet, unassuming and very open group of scientists whose work and credentials are transparent and easy to verify.

The Next Wave of Discovery

A world with an unlimited energy supply isn't hard to imagine; a greener world, with cheaper transportation, global connectivity, solutions to global health and food, not to mention the always elusive deep space travel - the possibilities are endless. So even as cold fusion seems to buzz around with a disreputable past the real deals aren't dismayed.

Scientists at the National Institute of Nuclear Physics (NINP) in Italy and NASA's Aeronautics Research Institute in the United States have made significant strides towards the fundamental research necessary to realise cool fusion. Italian scientist Francesco Celani at NINP has even made his research open and accessible for scrutiny, having demonstrated a modest success (5 to 10 watts energy) with the intention to grow slowly but firmly. And NASA is putting more resources in researching LENR reactors, solving each problem in a deliberate and patient fashion.

The need for this innovation is high, the time is running out and too many frivolous distractions are leaching from a noble cause, but as long as dedicated scientists continue to work tirelessly with determination and transparency, humanity will once again steal back the fire from the gods and set free the innovations of our imagination. 



A warm infinite fire for all mankind. Is that too much to ask?

And as nuclear and renewable sources of energy grow at a rate of 2.5 percent per year we can still expect 80 percent of energy from fossil fuels. Considering the past behaviour of human consumption and demand, this equation isn't likely to change, as we discussed in our story about oil consumption (Digit May 2013).

But until the first unit is in the hands of independent examiners we must hope that it works. And yet, unless Rossi dies under mysterious circumstances (we usually accept conspiracy as proof of reality, don't we?) or until scientists clearly explain the logic of his device, it's more rational to believe that while Fleischmann-Pons may have made a genuine scientific error, Rossi

conforms to all the scientific concepts - it actually allows for nuclear fusion reactions at significantly lower non-thermonuclear temperatures.

Theorised during the 1950s, it was applied by Luis Alvarez's team in 1956, resulting in the controlled generation of 5.5 MeV of energy. The success of this minor output was further investigated by John Jackson who was able to theoretically distill the practical applications of the process and its energy output.

Jackson's research outlined the practical limitations of the process as a commercial product and presented scientist's with a clear problem that needed solving to achieve success (read: alpha-sticking